

Math 140 Spring 2023 - Worksheet 2.2, 2.3

Section:

Name:

1. Find an equation of the line l tangent to the graph of $f(x) = \frac{\cos^2 x}{x}$ at the point $(0, 1)$.
Hint: Remember that the first step of obtaining a tangent line is finding the slope. You can use the fact that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.
2. Evaluate $\lim_{x \rightarrow 2} \left(\frac{\sqrt{x}}{x-4} - \frac{2}{x-4} \right)$. Reason why the rule $\lim_{x \rightarrow a} (f(x) + g(x)) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$ cannot be applied.
3. Evaluate $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos^2 x}{1 - \sin x}$.